

MODERN SYSTEMS ANALYSIS & DESIGN

# Managing the Information Systems Project

CHAPTER 3

# Learning Objectives

- 1 Explain how information systems projects are managed from start to finish.
- 2 Describe the skills an effective project manager needs.
- 3 Identify the project manager's activities across initiation, planning, execution, and closedown.
- 4 Explain critical path scheduling, and how to build Gantt charts and network diagrams.
- 5 Explain how project management software supports scheduling and tracking.

# Importance of Project Management

*Project management, **an important factor** in whether a systems development effort succeeds.*



## Meets Customer Expectations

Effective PM keeps deliverables aligned with what the customer asked for.



## Respects Budget & Time

It keeps the project inside the budget and schedule constraints set at the start.

*PM skills are difficult to master — and important to keep sharpening.*

# Deciding on Systems Projects



## System Service Request (SSR)

A standard form used to request or propose systems development work within an organization.



## Feasibility Study

An assessment of whether a requested system makes economic and operational sense for the organization.

EXAMPLE

# A System Service Request in Practice

| SYSTEM SERVICE REQUEST |                                                                   |
|------------------------|-------------------------------------------------------------------|
| REQUESTED BY           | Leonard Jackson, Purchasing                                       |
| DATE                   | October 1, 2026                                                   |
| DEPARTMENT             | Manufacturing / Purchasing                                        |
| TYPE OF REQUEST        | New System                                                        |
| PROBLEM STATEMENT      | Growing sales volume has outpaced what current systems can track. |
| LIAISON                | Chris Martin, Purchasing                                          |
| SPONSOR                | V.P. of Manufacturing                                             |

What goes into an SSR

- Who is requesting the system, and their contact information
- A clear statement of the business problem
- The liaison and sponsor who will support the project

FIGURE 3-2 — A Systems Priority Board reviews each SSR; limited time and resources mean not every request is approved.

# Managing the Information Systems Project



## Project

A planned undertaking of related activities aimed at a single objective — with a defined beginning and end.



## Project Management

The controlled process of initiating, planning, executing, and closing down a project.



## Project Manager

A systems analyst with the management and leadership skills to lead initiation, planning, execution, and closedown.



## Deliverable

The tangible end product produced at the close of an SDLC phase.

# Project Management Activities

*A project manager juggles many concurrent activities — the role is a hub, not a single track.*



# Phases of the Project Management Process



## PHASE 1

# Project Initiation

*Assess the project's size, scope, and complexity, and put initial procedures in place.*



**Initiation team**



**Customer relationship**



**Project initiation plan**



**Management procedures**



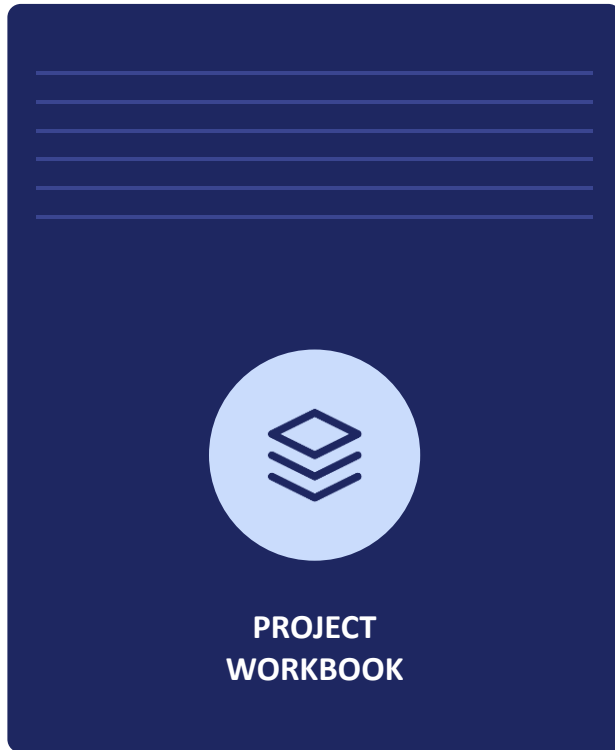
**Project management environment**



**Project workbook**

FIGURE 3-6 (RECREATED)

# The Project Workbook



- Project overview
- Scope statement & staffing plan
- Team correspondence
- Standards & procedures
- Change requests
- Original SSR & feasibility study
- Data & process descriptions
- Project schedule & charter
- Status reports

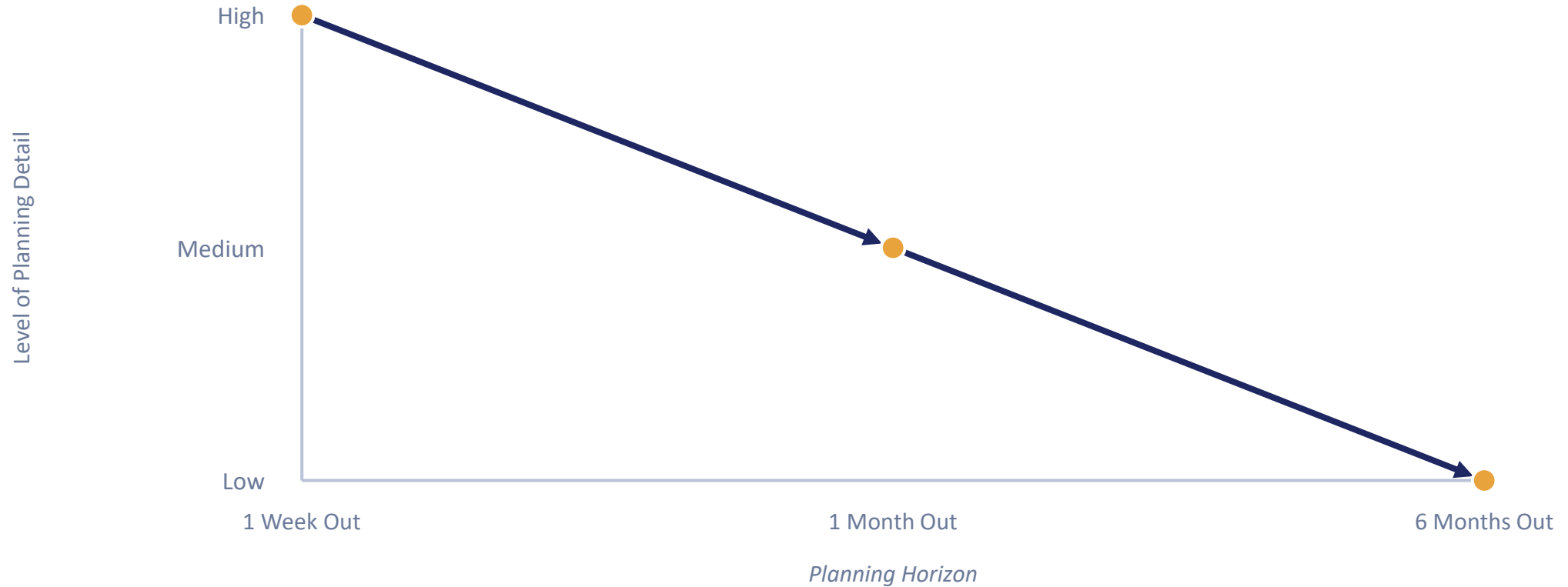
*Nine key documents, kept in both hard-copy and electronic form throughout the project.*

# Project Planning

*Define clear, discrete activities and the work needed to complete each one.*

- 1 Define scope, alternatives & feasibility
- 2 Divide the project into tasks
- 3 Estimate resource requirements
- 4 Draft a preliminary schedule
- 5 Develop a communication plan
- 6 Set standards & procedures
- 7 Identify and assess risk
- 8 Create a preliminary budget
- 9 Write a statement of work
- 10 Set the baseline project plan

# Planning Detail Over Time



*Planning detail should be high for the near term, with less detail the further out the plan reaches.*

# Components of Project Planning



## Statement of Work (SOW)

A “contract” between IS staff and the customer covering deliverables and time estimates.



## Baseline Project Plan (BPP)

Estimates of scope, benefits, schedules, costs, risks, and resource needs.



## Preliminary Budget

A cost-benefit analysis outlining planned expenses and revenues.



## Work Breakdown Structure

Dividing the project into manageable, logically ordered tasks and subtasks.



## Gantt Chart

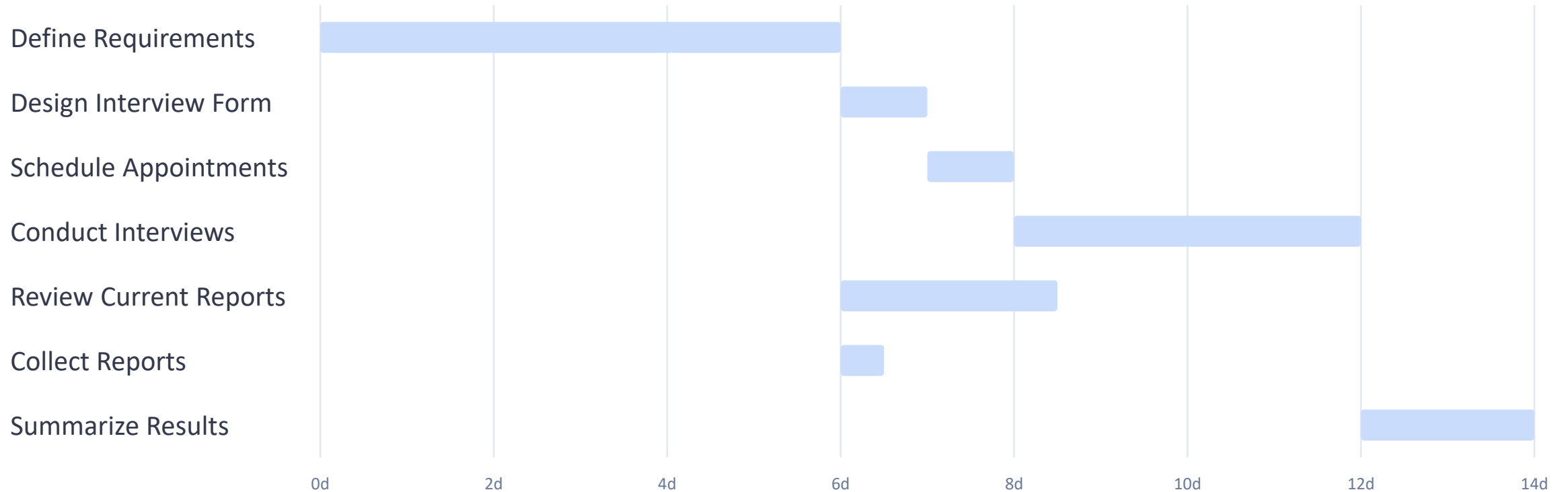
Horizontal bars represent task durations, laid out on a timeline.



## Network Diagram

Boxes and links represent dependencies between tasks.

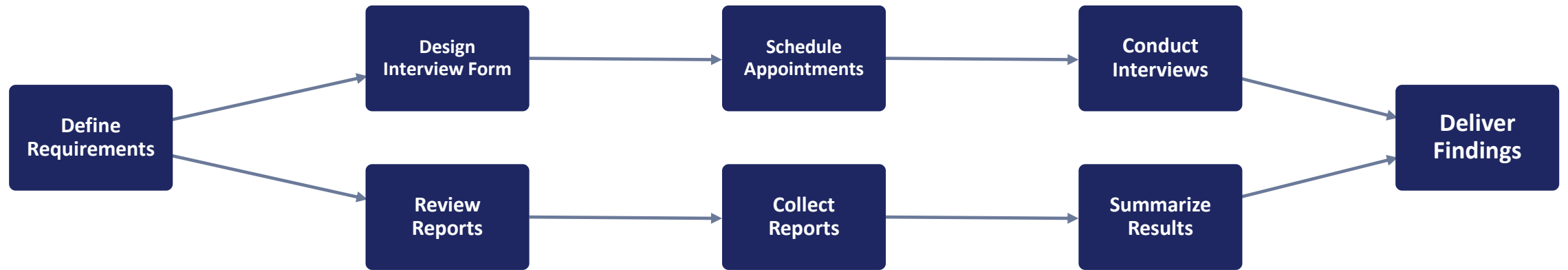
# Scheduling Diagrams: Gantt Chart



*Gantt chart showing project tasks, their durations, and how they overlap on a shared timeline (in days).*

**Special-purpose project management software is available for this.**

# Scheduling Diagrams: Network Diagram

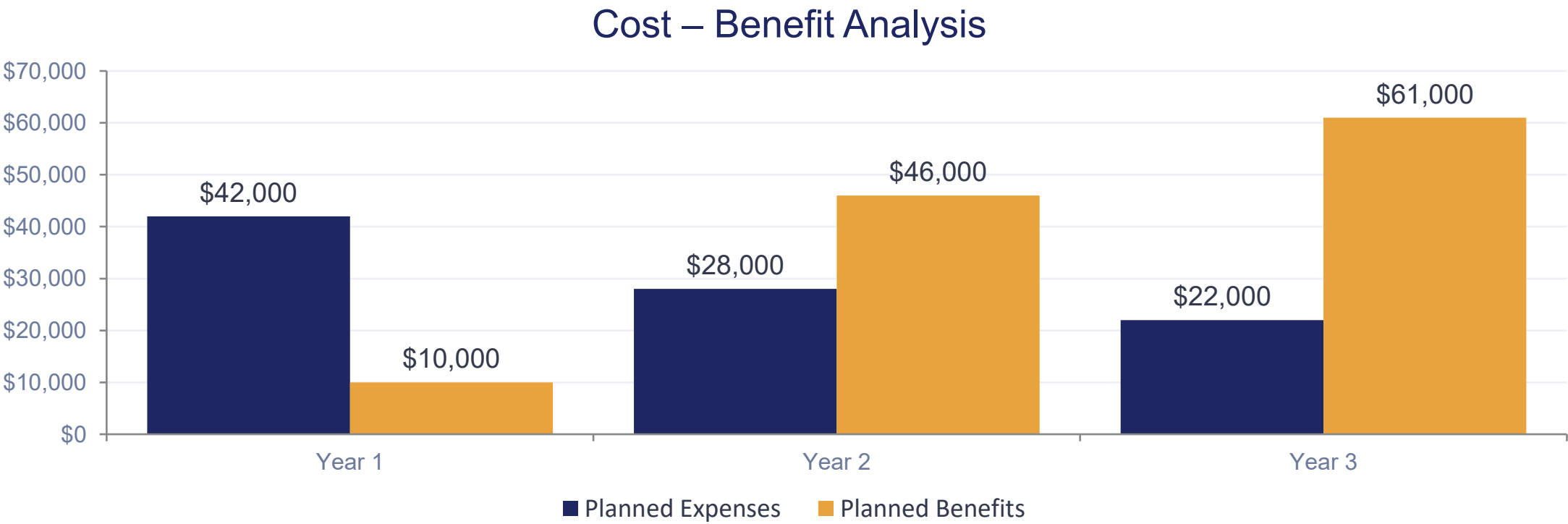


*Rectangles represent tasks; arrows show the sequence and dependencies between them.*

**Special-purpose project management software is available for this.**

FIGURE 3-15 (RECREATED)

# Preliminary Budget



*A financial cost-and-benefit analysis for a systems development project. Spreadsheet software is well suited to this.*

### PHASE 3

# Project Execution

*Plans created in the prior phases are put into action.*



**Execute the baseline project plan**



**Monitor progress against the baseline**



**Manage changes to the baseline plan**



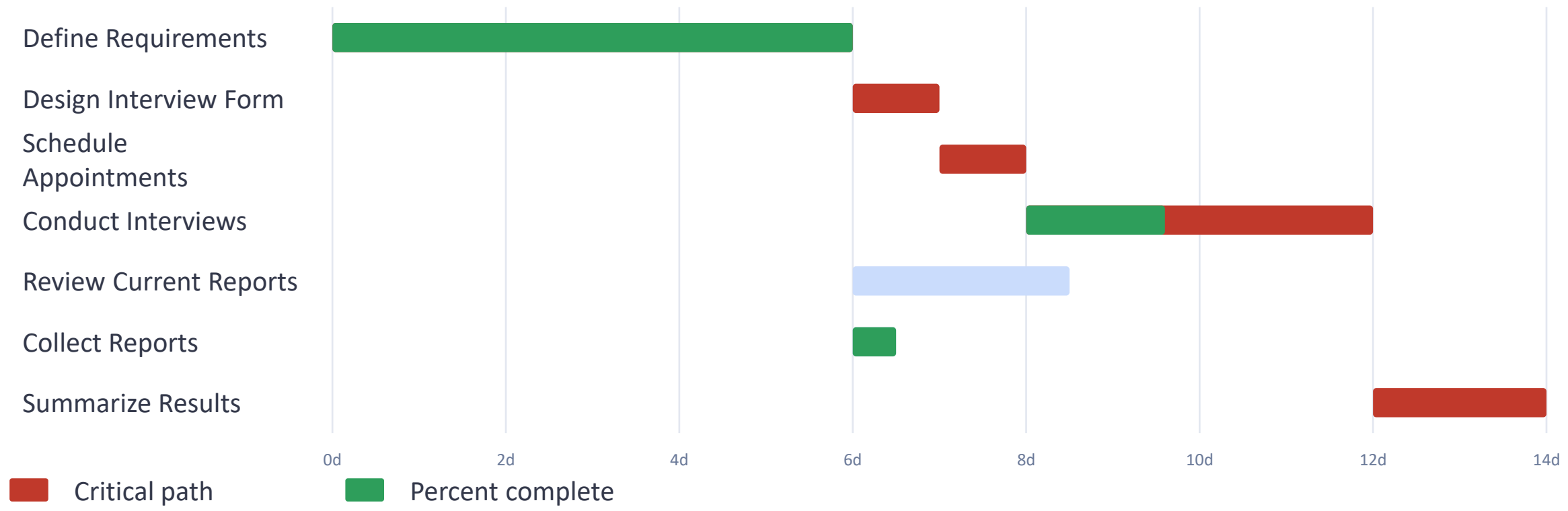
**Maintain the project workbook**



**Communicate project status**

FIGURE 3-17 (RECREATED)

# Monitoring Progress with a Gantt Chart



# Communication Methods

## High Formality

Project workbook · Status reports ·  
Specification documents · Meeting minutes

## Medium Formality

Meetings · Seminars & workshops · Memos

## Low Formality

Bulletin boards · Brown-bag lunches · Hallway discussions

## PHASE 4

# Project Closedown

*Bring the project to a formal end.*



**Close down the project**



**Conduct post-project reviews**



**Close the customer contract**

COMING UP

## Representing and Scheduling Project Plans



**Gantt Charts**



**Network  
Diagrams**



**PERT  
Calculations**



**Critical Path  
Scheduling**



**Project  
Management  
Software**

# Gantt Charts vs. Network Diagrams



## Gantt Charts

- Show task durations
- Show time overlap
- Show slack time within a bar's duration



## Network Diagrams

- Show task dependencies
- Show parallelism, not time overlap
- Show slack time inside the boxes

# Estimating Task Duration with PERT

*PERT — Program Evaluation Review Technique*

Combines an optimistic (o), pessimistic (p), and realistic (r) time estimate to produce one expected task duration.

$$ET = ( o + 4r + p ) / 6$$

**o**

Optimistic time

**r**

Realistic (most likely) time

**p**

Pessimistic time

**ET**

Expected task duration

# Example PERT Analysis

| Activity                   | o | r | p | Expected Time (ET) |
|----------------------------|---|---|---|--------------------|
| 1. Requirements Collection | 1 | 5 | 9 | 5.0                |
| 2. Screen Design           | 5 | 6 | 7 | 6.0                |
| 3. Report Design           | 3 | 6 | 9 | 6.0                |
| 4. Database Design         | 1 | 2 | 3 | 2.0                |
| 5. User Documentation      | 2 | 6 | 7 | 5.5                |
| 6. Programming             | 4 | 5 | 6 | 5.0                |
| 7. Testing                 | 1 | 3 | 5 | 3.0                |
| 8. Installation            | 1 | 1 | 1 | 1.0                |

***Estimated time calculations for the project, using  $ET = (o + 4r + p) / 6$ .***

# Critical Path Scheduling

*A scheduling technique in which the order and duration of a sequence of activities directly determines the project's completion date.*



## Critical Path

The shortest time in which the entire project can be completed — its longest chain of dependent tasks.



## Slack Time

The amount an activity can be delayed without pushing back the overall project.

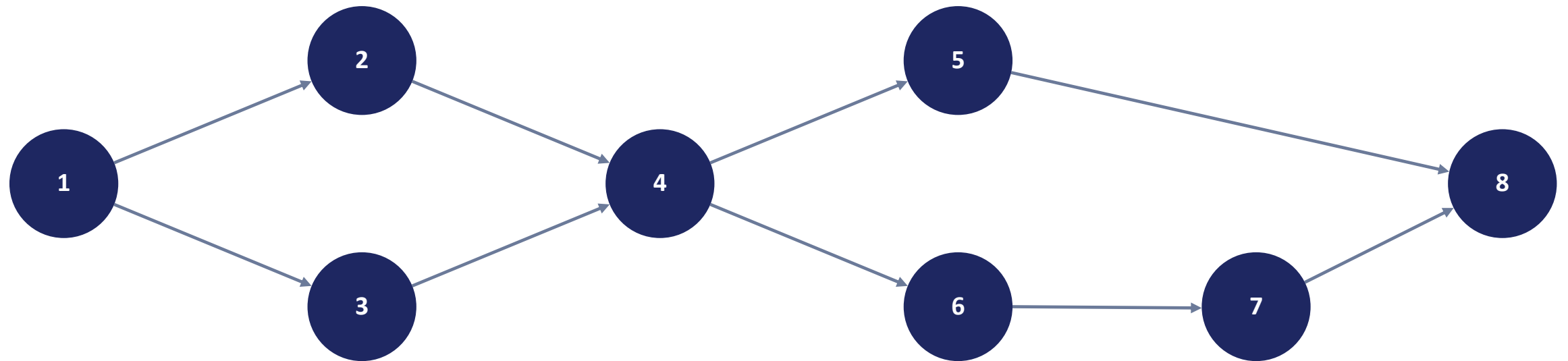
# Critical Path Example: Dependencies

| Activity                   | Preceding Activity |
|----------------------------|--------------------|
| 1. Requirements Collection | —                  |
| 2. Screen Design           | 1                  |
| 3. Report Design           | 1                  |
| 4. Database Design         | 2, 3               |
| 5. User Documentation      | 4                  |
| 6. Programming             | 4                  |
| 7. Testing                 | 6                  |
| 8. Installation            | 5, 7               |

*A preceding activity must finish before the specified activity can begin.*

This dependency table is the direct input to the network diagram on the next slide.

## Critical Path Example: Network Diagram



*Circles represent activities; arrows show the sequence between them.*

**Network diagram showing dependencies for the same eight activities.**

# Determining the Critical Path

1

## Forward Pass

Calculate each activity's earliest possible completion time by summing activity times along the longest path leading to it. This gives the total expected project time.

2

## Backward Pass

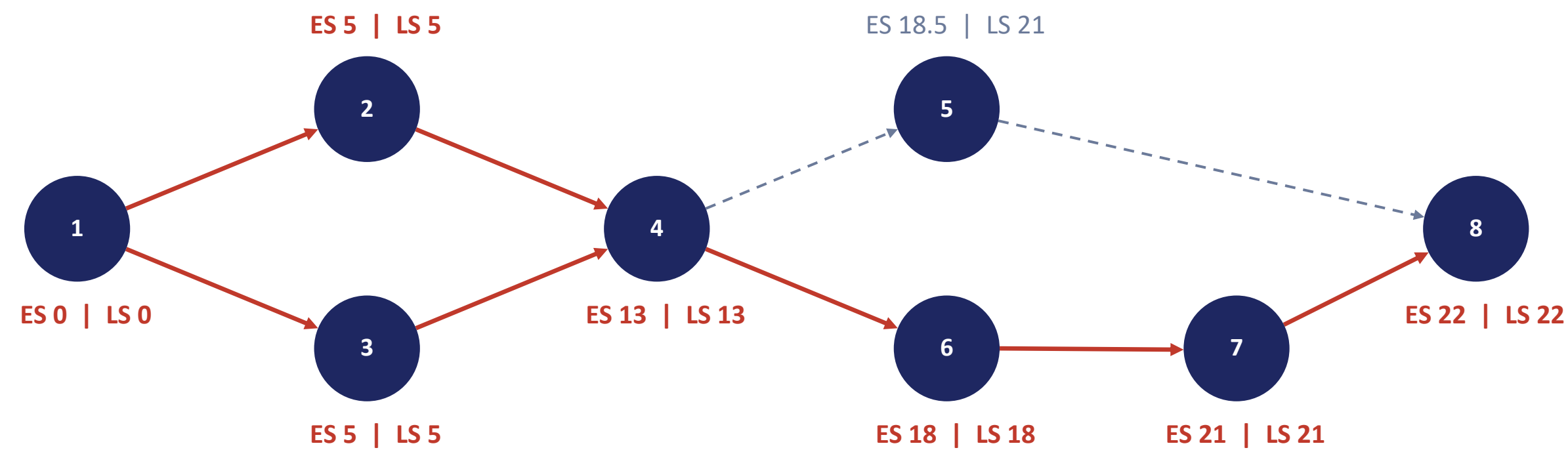
Calculate each activity's latest possible completion time by subtracting the activity times on the path that follows it from the total expected time. This gives each activity's slack.

3

## Identify the Path

The critical path is the chain of activities with zero slack — any delay there delays the whole project.

# Critical Path Calculation



Early and late times are calculated for each activity; Activity 5 can begin late without affecting the finish date.

## Critical Path Calculation (cont.)

| Activity | TE   | TL | Slack (TL – TE) | On Critical Path |
|----------|------|----|-----------------|------------------|
| 1        | 5    | 5  | 0               | ✓                |
| 2        | 11   | 11 | 0               | ✓                |
| 3        | 11   | 11 | 0               | ✓                |
| 4        | 13   | 13 | 0               | ✓                |
| 5        | 18.5 | 21 | 2.5             | —                |
| 6        | 18   | 18 | 0               | ✓                |
| 7        | 21   | 21 | 0               | ✓                |
| 8        | 22   | 22 | 0               | ✓                |

*Every activity except Activity 5 sits on the critical path — note its 2.5-day slack.*

## TOOLS

# Using Project Management Software

*Many powerful software tools exist to assist with project management — Microsoft Project is a common example.*



**Enter the project start date**



**Establish tasks and task dependencies**



**View project information as Gantt or network diagrams**

# Setting the Project Start Date

**PROJECT INFORMATION**



Project Start Date  
**March 3, 2027**

Scheduling mode: Auto · Calendar: Standard (5-day week)

Save

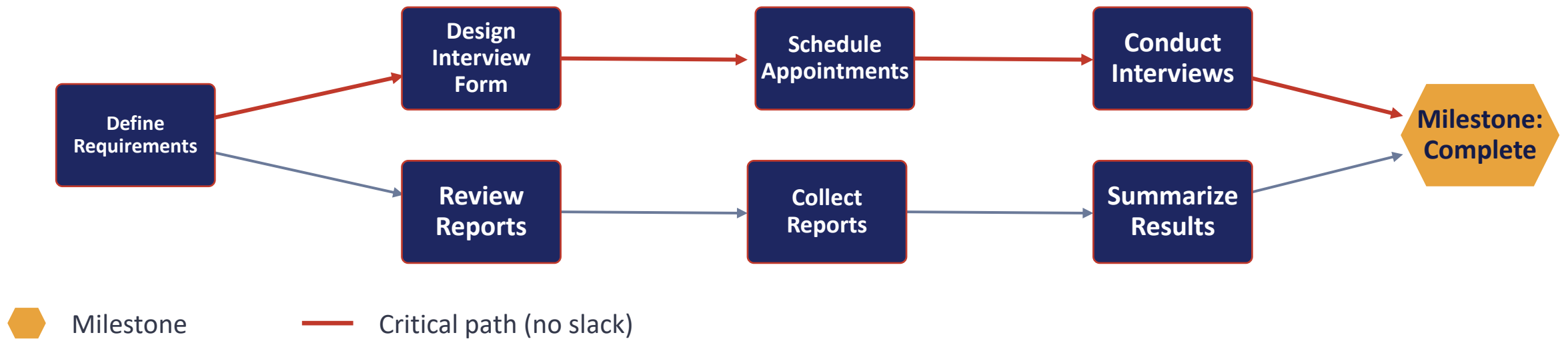
*Every schedule anchors to a starting date; the software calculates every other date from there.*

# Entering Tasks and Dependencies

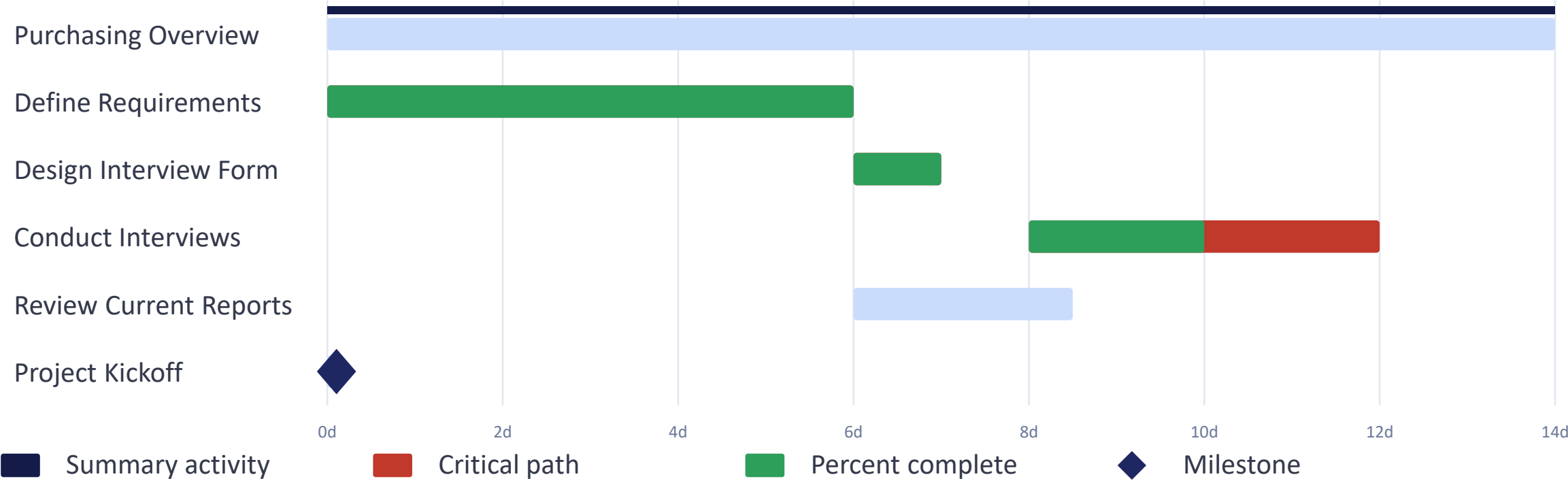
| Task Name                 | Duration | Predecessors |
|---------------------------|----------|--------------|
| 1. Define Requirements    | 6 days   | —            |
| 2. Design Interview Form  | 1 day    | 1            |
| 3. Schedule Appointments  | 1 day    | 2            |
| 4. Conduct Interviews     | 4 days   | 3            |
| 5. Review Current Reports | 2.5 days | 1            |
| 6. Collect Reports        | 0.5 days | 1            |
| 7. Summarize Results      | 2 days   | 4, 6         |

*Once tasks, durations, and predecessors are entered, the software builds the Gantt and network views automatically.*

# Viewing a Network Diagram in Software



# Viewing a Gantt Chart in Software



# Summary

- ✓ How information systems projects are managed from initiation through closedown.
- ✓ The skills an effective project manager needs.
- ✓ The project manager's activities across each of the four PM phases.
- ✓ Critical path scheduling, and how to build Gantt charts and network diagrams.
- ✓ How project management software supports scheduling and tracking.